CS & IT ENGINEERING



Algorithms

Analysis of Algorithms



Lecture No.- 01

By- Aditya sir

Topics to be Covered



Topic Schedule

Topic Outcomes

Topic Intro to Algorithms



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

Join the telegram channel



"Aditya Jain Sir PW"

Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe

Syllabus : Algo



Topic : Lecture Schedule

Ch 1. Analysis of Algorithms

1. Algorithm Concept and Lifecycle
2. Analysis of Algorithms
3. Methodology & Types of Analysis
4. Asymptotic Notations ☆☆
5. Framework for Analysing Recursive Algorithms
6. Apriori analysis of Non-Recursive Algorithms
7. Analysing Loops
8. Space Complexity
9. Mathematical Background ☆

DAA

2 Design & 1 Analysis of Algorithms

Time & Space Complexity.

Design

1) Divide & Conquer

2) Greedy Algo

3) Dynamic Programming



Topic : Lecture Schedule

2. Divide & Conquer (Design Strategies) → British Rule

1. General Method → Mindset
2. Max-Min Problem
3. Merge Sort
4. Binary Search
5. Quick Sort

V.V.Imp

6. Matrix Multiplication → Strassen's MM
7. Long Integer Multiplication (LIM) → Bonus
8. Master Method for D and C Recurrences
9. Recursion Tree



Topic : Lecture Schedule

3. Greedy Method
   1. General Method
   2. Knapsack Problem
   3. Job Sequencing with Deadlines
   4. Optimal Merge Patterns
   4. Huffman Coding
   5. Minimum Cost Spanning Trees
      1. Prims Method
      2. Kruskal's Method
   6. Dijkstra's Shortest Paths Problem

fractional

methodology

Merge Algo

MCST

SSSP

PYQs + DPPs + WT + Questions in class



Topic : Lecture Schedule

4. Dynamic Programming (DP)
   1. The Framework
   2. Difference between DP, Greedy Method and Divide and Conquer
   3. Multistage Graphs
   4. Travelling Salesperson Problem
   5. Binary Knapsack Problem
   7. Bellman-Ford Single Source Shortest Paths
   8. Longest Common Subsequence
   10. Matrix Chain Multiplication
   11. Optimal Cost Binary Search Tree



TSP → 0/1 Knapsack

All Pairs Shortest Paths → APSP → Floyd Warshall

LCS

sum of Subsets → SOS

MCM

Bonus



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5. Graph Algorithms

1. Representation of Graphs
2. Graph Traversals

DFS

- 5.2.1 Undirected Connected Graphs
- 5.2.2 Undirected Disjoint Graphs: DFT
- 5.2.3 Directed Graphs & Types of Edges ★★
- 5.2.4 DAG

BFS

- 5.2.5 FIFO BFS ★★
- 5.2.6 LIFO BFS

Bonus

Directed / Undirected
wt / unwt

Stack ⟹ LIFO

Directed Acyclic Graph.

Queue → FIFO

Queue default



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6. Heap Algorithms

1. Operations : Create, Insert, Delete, Modify
2. Applications : Heapsort

Min / Max

GBG

6-7 Ques

65

CS



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7. Sets → Kruskal Algo ☆☆
   1. Representations
   2. Operations → Union( ) + find( )



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8. Sorting Algorithms ☆☆
    1. Basic terminology
    2. Methods
        1. Bubble Sort
        2. Selection Sort
        3. Insertion Sort
        4. Radix Sort

→ Merge
→ Quick
} D n Q

→ Heap → Heaps



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9. Backtracking & Branch- Bound ⟶ Bonus

↳ Last



Topic : Lecture Schedule

Reference Books:

1. Introduction to Algorithm → Cormen
2. Fundamentals of Algorithm → Sahni

Bible of Algo

Algo : 8 - 10 on an Avg



Topic : Lecture Schedule

Scope/ Outcomes

GATE, TIFR, ISRO, ~~BASIC~~ ... BARC . . . . .

Placement → Product/service based

Google	Infosys
Microsoft	WIPRO
	Accenture
	TCS

☆ Coding Tests

1st Interview

SDE 1st Dec 7:00 AM



Topic : Lecture Schedule

Pre-Requisites

1. Data Structures Fundamental
   - [Stacks, Ques, Tree…]
2. Programming Fundamental
   - If-else
   - Loops
3. Basic Maths

   Series (AP, GP) → Revision.

if else loops

GATE

8LPA

30th Jan → Amazon

3hrs (2 hrs)

Coding Test (200+)

3rd Feb → GATE ✓

9th Feb → Interview.

✓

7 months

Algorithm ?



Topic : Lecture Schedule

Algorithm:-

An Algorithm is a collection of finite number of instruction to solve a given problem.

These instruction are fundamental and should follow a proper sequence.

a*c+b-d

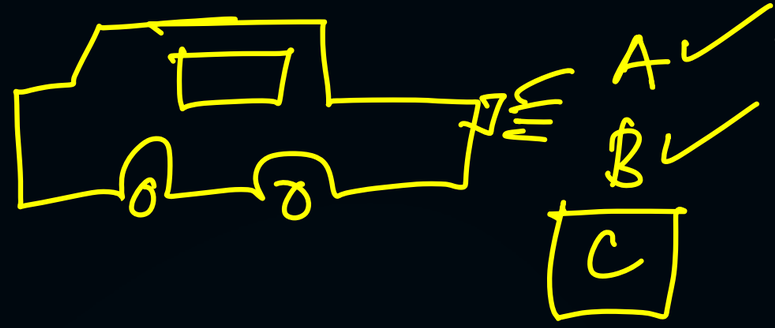
atomic

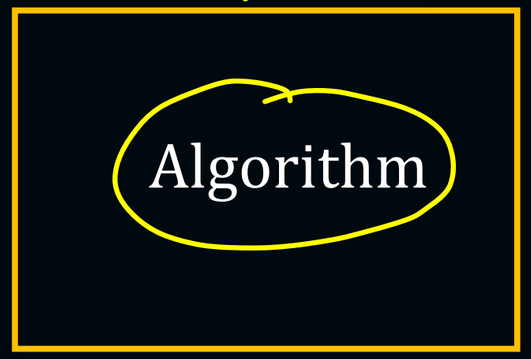
Step-by-Step



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BSC





Input (i/p) → Output (o/p)

[An algorithm may take 0 or more i/ps]

may or may not take an i/p.

[Must always produce at least one output]

Black box

```
void greet()
{
    print("Hello AJ")
}
```



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Program vs Algo

Python:- for i in range (0, n+1)

C++:- For (i =0, I < =0; i + +)

C
C++
Java, Python

Program→ Algorithm implemented using some programing language

Algorithm → Pseudo code- for i → o : n

for i = 0 to n

pseudo Code / Logic

(Set of sequential rules /statement /instructions)

For i ⟶ 1 to n ⟶ for i in range (0, n+1)

⟶ for (i = 0; i <= n; i++)



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Algorithm Lifecycle

College

Problem statement → Requirements /constraints → Logic / Design Algorithm

Analysis (Time complexity) (efficiency)
To solve problem in the most efficient way

Coding programming (To run it own machine)

Testing

Debug

check its correctness

Completed



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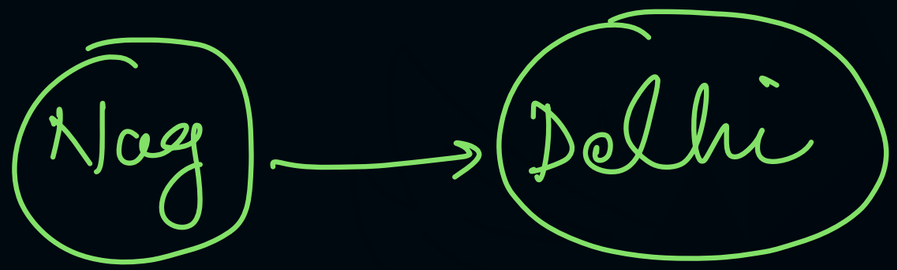
```
Algo AJSir (  )   → 0 inputs
{
        print (“Hello students!”)
  }
        Algo AJsir2(  )
 {
        return 100
   }
```



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Analysis of Algorithm:-

1. Why to analysis ?→ Compare two Algo. or more
2. What to analysis?→ on the basis of time/space {~~Resudece consume point~~} ★ Resource Consumption.
3. How to analysis?





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#Q. Why to Analysis ?

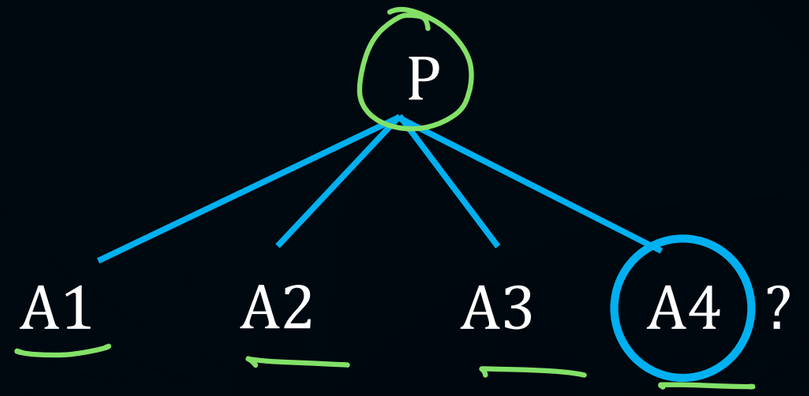
Algo / Soln

To compare and decide which ~~sudation~~ is the best among different available solution (Algo.)

Problem :- Nagpur → Delhi

"Best"

1. Walk → Soln. (A1)
2. Bike /Car→ (A2)
3. Train→ (A3)
4. Flight→ (A4)

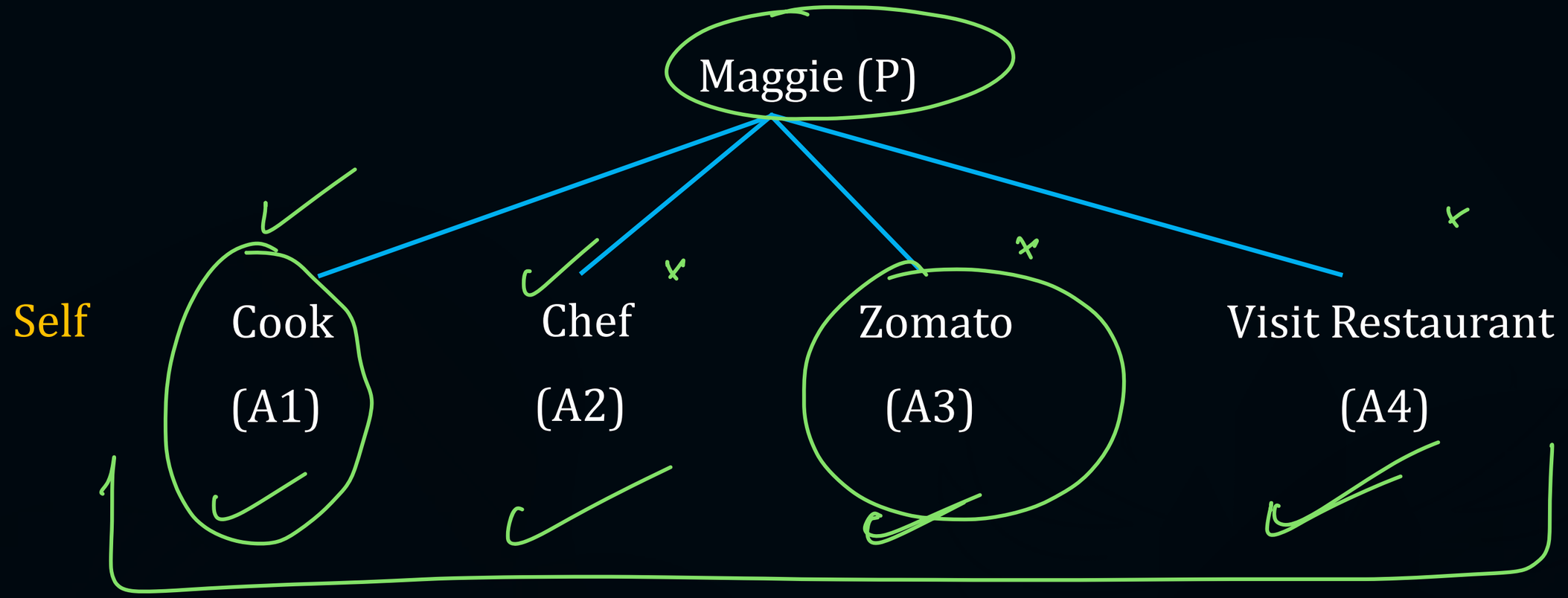


Time ↓ cost ↑ (Money)





Topic : Lecture Schedule





Topic : Lecture Schedule

#Q. What to Analysis ?

Analysis the consumption of Resources

Resources – Time, space /memory → GATE

Money, Internal Bandwidth, etc.

Number of programmers.

other metrics



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#Q. How to Analysis ?

How to Analysis ?

1) Aposteriori

1. Aposteriori Analysis

Analyzing After Implementation of Algo.

Code → Run

2. Apriori Analysis

Analyzing before implementation of Algo.

no Code just logic



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1. Apostcriori Analysis

Advantages

- Given the exact measurement of time units.

High level long → User Programm friendly

Low level long → Machine friendly

Disadvantage

- Platform dependent

Software → OS (Linux, windows MAC)

Hardware → (Process /CPU memory (RAM))

Programming language dependent

Hight → Slower

Low → 0/1 faster

Python, C++, C, Bash

Algo1

↓

efficient (3 sec)

↓

(i20, windows 15)

↓

10 sec

Algo2

↓

inefficient (7 sec)

↓

(i3, windows 7)

↓

2 sec

2) Apriori Analysis :-

→ Before implementation

Disadv :- → approx

Adv : 1) Platform independent
2) Easy understand
3) Testable.

H.W

?

How Compare ?



Summary

1) Syllabus (detail)

2) Pre-requisite & Resources

3) Intro to Analysis

" 6:30 PM "



THANK - YOU